

Segmentation

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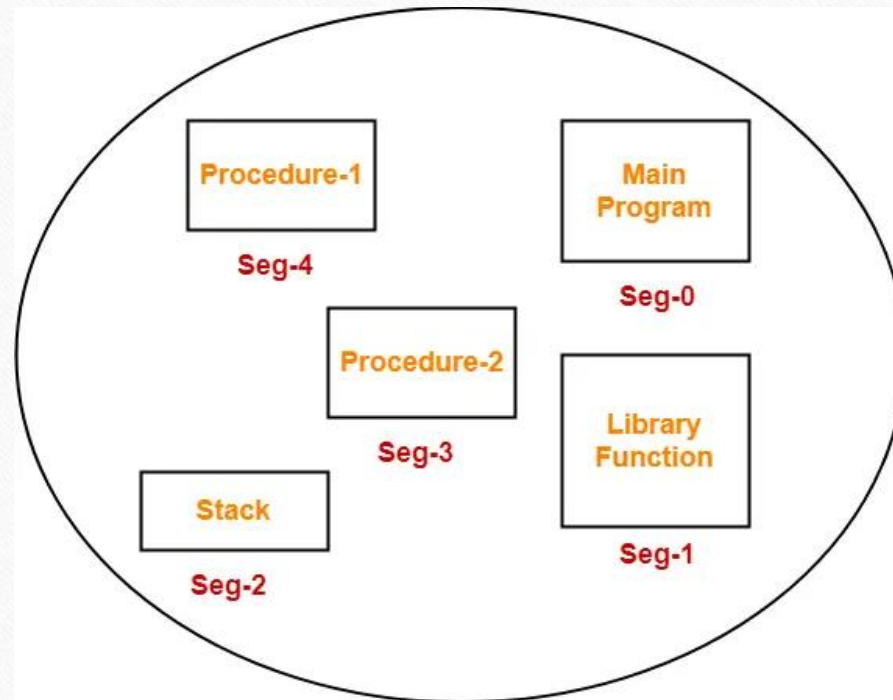
Segmentation

- A process is divided into Segments. The chunks that a program is divided into which are not necessarily all of the exact sizes are called segments.
- Segmentation gives the user's view of the process which paging does not provide. Here the user's view is mapped to physical memory.
- Like Paging, Segmentation is another non-contiguous memory allocation technique.
- In segmentation, process is not divided blindly into fixed size pages.
- Rather, the process is divided into modules for better visualization.

Characteristics

- Segmentation is a variable size partitioning scheme.
- In segmentation, secondary memory and main memory are divided into partitions of unequal size.
- The size of partitions depend on the length of modules.
- The partitions of secondary memory are called as **segments**.

Segments



Segment table

- Segment table is a table that stores the information about each segment of the process.
- It has two columns.
- First column stores the size or length of the segment.
- Second column stores the base address or starting address of the segment in the main memory.
- Segment table is stored as a separate segment in the main memory.
- Segment table base register (STBR) stores the base address of the segment table.

Segment table

	Limit	Base
Seg-0	1500	1500
Seg-1	500	6300
Seg-2	400	4300
Seg-3	1100	3200
Seg-4	1200	4700

Segment Table

Translating Logical Address into Physical Address

- CPU always generates a logical address.
- A physical address is needed to access the main memory.
- Following steps are followed to translate logical address into physical address-

Step-01:

- CPU generates a logical address consisting of two parts-
 1. Segment Number
 2. Segment Offset
- Segment Number specifies the specific segment of the process from which CPU wants to read the data.
- Segment Offset specifies the specific word in the segment that CPU wants to read.

Translating Logical Address into Physical Address

Step-02:

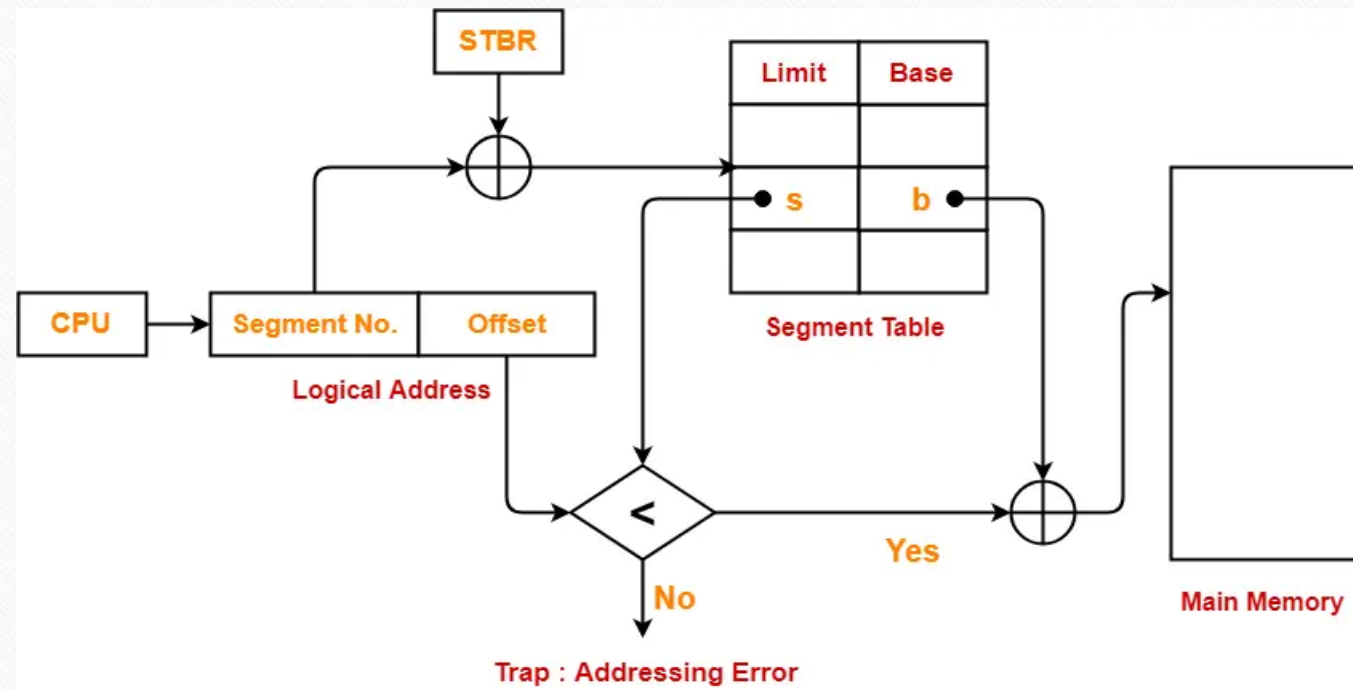
- For the generated segment number, corresponding entry is located in the segment table.
- Then, segment offset is compared with the limit (size) of the segment.

Case-01: Segment Offset $\geq$ Limit

- If segment offset is found to be greater than or equal to the limit, a trap is generated.

Case-02: Segment Offset $<$ Limit

- If segment offset is found to be smaller than the limit, then request is treated as a valid request.
- The segment offset must always lie in the range $[0, \text{limit}-1]$,
- Then, segment offset is added with the base address of the segment.
- The result obtained after addition is the address of the memory location storing the required word.



Translating Logical Address into Physical Address

Advantages and disadvantages

The advantages of segmentation are-

- It allows to divide the program into modules which provides better visualization.
- Segment table consumes less space as compared to **Page Table** in paging.
- It solves the problem of internal fragmentation.

The disadvantages of segmentation are-

- There is an overhead of maintaining a segment table for each process.
- The time taken to fetch the instruction increases since now two memory accesses are required.
- Segments of unequal size are not suited for swapping.
- It suffers from external fragmentation as the free space gets broken down into smaller pieces with the processes being loaded and removed from the main memory.